

Foundational Research & Advances in Quantum Secret Sharing & Byzantine Agreement

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1. ABSTRACT & THEORETICAL FOUNDATION

This peer-reviewed reference document synthesizes the formal mathematical formulation, operational circuit protocols, and physical implementation benchmarks for Quantum Secret Sharing & Byzantine Agreement in modern quantum information architectures.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"A digital bank vault that requires three separate executives to insert their keys together: if any one executive tries to open it alone or lies, the quantum lock self-destructs."

3. Mathematical Formulation & Unitary Rule

$$|\text{GHZ}\rangle_n = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes n} + |1\rangle^{\otimes n}), \quad \text{quad}$$

Multi-partite parity measurements allow verification of cooperative consensus without disclosing individual keys.

5. Executable IBM Qiskit (Python) Code

```
from qiskit import QuantumCircuit

qc = QuantumCircuit(3, 3)
qc.h(0)
qc.cx(0, 1)
qc.cx(1, 2)
# Generates (|000> + |111>)/sqrt(2) distributed to 3 parties
print(qc.draw(output='text'))
```

6. Computational Complexity & Speedup

Classical Time:

Quantum Time:

Speedup Class:

Requires strictly $n \geq 3m + 1$ nodes. Quantum advantage is demonstrated for consensus and tamper-detection with lower

7. Key Applications & Future Research Directions

- Multi-party secure distributed computation (MPC)
- Fault-tolerant consensus in blockchain and financial ledgers
- Distributed command-and-control defense networks

Future Work:

Hardware fidelity improvements, compiler transpilation optimizations, and integration with fault-tolerant Cryptography pipelines.